Module 08 | Backpropagation Fundamentals

Problem Statement (Rephrased from Practice Problems)

Design and analyze a **fully connected neural network** with:

- **3 input neurons**
- **3 hidden layers**
- **1 output neuron**

You may choose the number of neurons in each hidden layer and specify the activation function used in each layer.

Complete the following tasks:

1. **Define the Network Architecture**
   - Clearly specify the number of neurons in each layer.
   - Specify the activation function used in each hidden and output layer.
   - Define the dimensions of all weight matrices and bias vectors.
2. **Initialize the Parameters**
   - Initialize all weights and biases with small numerical values.
   - Clearly list the initialized values.
3. **Perform the Forward Pass**
   - Choose one training example.
   - Perform the complete forward pass through the network.
   - Calculate the output step by step, showing the intermediate values at every layer.
4. **Define the Loss Function**
   - Define a suitable loss function, such as **Mean Squared Error (MSE)**.
   - Calculate the loss for the chosen training example.
5. **Perform the Complete Backward Pass and Compute All Gradients**
   - Derive the backpropagation equations using the chain rule.
   - Propagate the error from the output layer backward through all three hidden layers.
   - Compute the gradient for every weight and bias.

- Show the resulting gradient matrices and vectors.
- Clearly explain the mathematical reasoning at each layer.

6. **Perform One Parameter Update**
   - Choose a suitable learning rate.
   - Perform one complete **gradient descent** update.
   - Calculate and show the updated weights and biases for every layer.

The final solution should contain all intermediate calculations so that the entire **forward pass → loss calculation → backward pass → parameter update** can be followed step by step.

My Solution

TASK 01. Define the Network Architecture

Input Layer

Number of Neurons: 3
So, any input sample consists of 3 feature values.

An Input Sample:

$$x \equiv \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

, a 3×1 matrix, that is, a 3×1 tensor.

x can be thought as **the output** of the input layer, that is, in general notation, $A^{(0)} = x$.

Hidden Layer 1

Number of Neurons: 2, with each neuron having 3 weights and a bias.

Vectorization of Weights:

$$W^{(1)} \equiv \begin{bmatrix} w_{11}^{(1)} & w_{12}^{(1)} & w_{13}^{(1)} \\ w_{21}^{(1)} & w_{22}^{(1)} & w_{23}^{(1)} \end{bmatrix}$$

, a 2×3 matrix/tensor. Each row represents a neuron in this layer. And $w_{ij}^{(l)}$ corresponds to the connection between i-th neuron of this layer l, which is 1 in this case, and j-th neuron of the previous layer.

Vectorization of Biases:

$$B^{(1)} \equiv \begin{bmatrix} b_1^{(1)} \\ b_2^{(1)} \end{bmatrix}$$

, a 2×1 matrix/tensor. b_i is the bias parameter of i-th neuron of this layer.

Activation Function:

$$\mathrm{ReLU}(z) = \begin{cases} z, \text{if } z \geq 0 \\ 0, \text{if } z < 0 \end{cases}$$

Linear Weighted Sum:

$$\begin{aligned} Z^{(1)} &\equiv W^{(1)} A^{(0)} + B^{(1)} \\ &\equiv W^{(1)} x + B^{(1)} \end{aligned}$$

Output:

$$A^{(1)} \equiv \mathrm{ReLU}(Z^{(1)})$$

Hidden Layer 2

Number of Neurons: 3, with each neuron having 2 weights and a bias.

Vectorization of Weights:

$$W^{(2)} \equiv \begin{bmatrix} w_{11}^{(2)} & w_{12}^{(2)} \\ w_{21}^{(2)} & w_{22}^{(2)} \\ w_{31}^{(2)} & w_{32}^{(2)} \end{bmatrix}$$

, a 3×2 matrix/tensor. Each row represents a neuron in this layer. And w_{ij} corresponds to the connection between i-th neuron of this layer and j-th neuron of the previous layer.

Vectorization of Biases:

$$B^{(1)} \equiv \begin{bmatrix} b_1^{(2)} \\ b_2^{(2)} \\ b_3^{(2)} \end{bmatrix}$$

, a 3×1 matrix/tensor. b_i is the bias parameter of i-th neuron of this layer.

Activation Function:

$$\text{ReLU}(z) = \begin{cases} z, \text{if } z \geq 0 \\ 0, \text{if } z < 0 \end{cases}$$

Linear Weighted Sum:

$$Z^{(2)} \equiv W^{(2)} A^{(1)} + B^{(2)}$$

Output:

$$A^{(2)} \equiv \text{ReLU}(Z^{(2)})$$

Hidden Layer 3

Number of Neurons: 2, with each neuron having 3 weights and a bias.

Vectorization of Weights:

$$W^{(3)} \equiv \begin{bmatrix} w_{11}^{(3)} & w_{12}^{(3)} & w_{13}^{(3)} \\ w_{21}^{(3)} & w_{22}^{(3)} & w_{23}^{(3)} \end{bmatrix}$$

, a 2×3 matrix/tensor. Each row represents a neuron in this layer. And w_{ij} corresponds to the connection between i-th neuron of this layer and j-th neuron of the previous layer.

Vectorization of Biases:

$$B^{(3)} \equiv \begin{bmatrix} b_1^{(3)} \\ b_2^{(3)} \end{bmatrix}$$

, a 3×1 matrix/tensor. b_i is the bias parameter of i-th neuron of this layer.

Activation Function:

$$\mathrm{ReLU}(z) = \begin{cases} z, \text{if } z \geq 0 \\ 0, \text{if } z < 0 \end{cases}$$

Linear Weighted Sum:

$$Z^{(3)} \equiv W^{(3)} A^{(2)} + B^{(3)}$$

Output:

$$A^{(3)} \equiv \mathrm{ReLU}(Z^{(3)})$$

Output Layer

Number of Neurons: 1, with 2 weights and a bias.

Vectorization of Weights:

$$W^{(4)} \equiv \begin{bmatrix} w_{11}^{(4)} & w_{12}^{(4)} \end{bmatrix}$$

, a 1×2 matrix/tensor. Each row represents a neuron in this layer. And w_{ij} corresponds to the connection between i-th neuron of this layer and j-th neuron of the previous layer.

Vectorization of Biases:

$$B^{(3)} \equiv \begin{bmatrix} b_1^{(4)} \end{bmatrix}$$

, a 3×1 matrix/tensor. b_i is the bias parameter of i-th neuron of this layer.

Activation Function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Linear Weighted Sum:

$$Z^{(4)} \equiv W^{(4)} A^{(3)} + B^{(4)}$$

Output:

$$\hat{Y} \equiv A^{(4)} \equiv \sigma(Z^{(4)})$$

TASK 02. Initialize the Parameters

Hidden Layer 1

$$W^{(1)} = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix}$$

$$B^{(1)} = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$$

Hidden Layer 2

$$W^{(2)} = \begin{bmatrix} 0.1 & 0.2 \\ 0.1 & 0.2 \\ 0.1 & 0.2 \end{bmatrix}$$

$$B^{(2)} = \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix}$$

Hidden Layer 3

$$W^{(3)} = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix}$$

$$B^{(3)} = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$$

Output Layer

$$W^{(4)} = \begin{bmatrix} 0.1 & 0.2 \end{bmatrix}$$

$$B^{(4)} = \begin{bmatrix} 0.1 \end{bmatrix}$$

TASK 03. Perform the Forward Pass

Choosing a Training Example

$$x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Hidden Layer 1

$$\begin{aligned} Z^{(1)} &\equiv W^{(1)}x + B^{(1)} \\ &= \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} \\ &= \begin{bmatrix} 0.7 \\ 0.8 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A^{(1)} &\equiv \mathrm{ReLU}(Z^{(1)}) \\ &= \mathrm{ReLU}\Big(\begin{bmatrix} 0.7 \\ 0.8 \end{bmatrix} \Big) \\ &= \begin{bmatrix} 0.7 \\ 0.8 \end{bmatrix} \end{aligned}$$

Hidden Layer 2

$$
\begin{aligned}
Z^{(2)} &\equiv W^{(2)}A^{(1)} + B^{(2)} \\
&= \begin{bmatrix} 0.1 & 0.2 \\ 0.1 & 0.2 \\ 0.1 & 0.2 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.8 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} \\
&= \begin{bmatrix} 0.33 \\ 0.43 \\ 0.53 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
A^{(2)} &\equiv \mathrm{ReLU}(Z^{(2)}) \\
&= \mathrm{ReLU}\left(\begin{bmatrix} 0.33 \\ 0.43 \\ 0.53 \end{bmatrix} \right) \\
&= \begin{bmatrix} 0.33 \\ 0.43 \\ 0.53 \end{bmatrix}
\end{aligned}
$$

Hidden Layer 3

$$
\begin{aligned}
Z^{(3)} &\equiv W^{(3)}A^{(2)} + B^{(3)} \\
&= \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 0.33 \\ 0.43 \\ 0.53 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} \\
&= \begin{bmatrix} 0.378 \\ 0.478 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
A^{(3)} &\equiv \mathrm{ReLU}(Z^{(3)}) \\
&= \mathrm{ReLU}\left(\begin{bmatrix} 0.378 \\ 0.478 \end{bmatrix} \right) \\
&= \begin{bmatrix} 0.378 \\ 0.478 \end{bmatrix}
\end{aligned}
$$

Output Layer

$$
\begin{aligned}
Z^{(4)} &\equiv W^{(4)} A^{(3)} + B^{(4)} \\
&= \begin{bmatrix} 0.1 & 0.2 \end{bmatrix} \begin{bmatrix} 0.378 \\ 0.478 \end{bmatrix} + \begin{bmatrix} 0.1 \end{bmatrix} \\
&= \begin{bmatrix} 0.2334 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
\hat{Y} \equiv A^{(4)} &\equiv \sigma(Z^{(4)}) \\
&= \sigma\Big(\begin{bmatrix} 0.2334 \end{bmatrix} \Big) \\
&\approx \begin{bmatrix} 0.558 \end{bmatrix}
\end{aligned}
$$

TASK 04. Define the Loss Function and Calculate the Loss

I have already used the sigmoid function in the output layer, so, yes, I am assuming this is a Binary Classification Problem.

I will use **Binary Cross-Entropy (BCE)** as my Loss Function.

$$
L \equiv \mathrm{BCE} \equiv -\Big[y \log \hat{y} + (1 - y) \log{(1 - \hat{y})} \Big]
$$

Calculating the Loss

For my input sample x, I will assume my true label is $y = 1$. And from **Forward Pass**, $\hat{y} = 0.558$.

$$
L = -1 \times \log 0.558 = -\log 0.558 \approx 0.253
$$

TASK 05. Perform the Complete Backward Pass and Compute All Gradients

Proud Documentation of My Satisfying Exploration

I belive my mentor wanted me to calculate the gradients of all scalers w and b, individually. He aimed for us to understand **what is truly happening** *under the 'magical'* **Autodifferentiation** provided by frameworks.

But, as a Linear Algebra fan, I couldn't resist **Vectorization**, which is a must for efficient implementation of backpropagation.

In one of my last git commits, I started calculating gradients in vectorized approach, keeping the **'scaler' reality** parallelly in my head. Soon, I noticed 'head' is not enough. So, I started scribbling on paper, of which I am so proud now. I have decided to keep it as a part of this repo. See pen-paper.pdf. I truly learned a lot though this.

I also consulted with Gemini. See my chats with Gemini: Matrix Multiplication Variants Explained, Understanding Jacobian Matrix and Vectorized Backpropagation.

Now, I know what is truly happening at scalar level as well as I have clarity on vectorized formulas. I am happy. I believe my mentor would be too. I am not lazy. I am differently industrious.

Summary of My Exploration

The following summary is tied to my definition of architecture in TASK 01. Define the Network Architecture.

Let's calculate the derivative of loss L with respect to a (scaler) weight $w_{ij}^{(l)}$, corresponding to the connection between i-th neuron of layer l and j-th neuron of the previous layer.

$$\begin{aligned}\frac{\partial L}{\partial w_{ij}^{(l)}} &= \frac{\partial L}{\partial a_i^{(l)}} \cdot \frac{\partial a_i^{(l)}}{\partial z_i^{(l)}} \cdot \frac{\partial z_i^{(l)}}{\partial w_{ij}^{(l)}} \\ &= \left(\sum_{k=1}^{n^{(l+1)}} \left(\frac{\partial L}{\partial z_k^{(l+1)}} \cdot \frac{\partial z_k^{(l+1)}}{\partial a_i^{(l)}} \right) \right) \cdot \frac{\partial a_i^{(l)}}{\partial z_i^{(l)}} \cdot \frac{\partial z_i^{(l)}}{\partial w_{ij}^{(l)}}\end{aligned}$$

As expressed by 'Back' in '**Back**propagation', we have to combine derivatives from left.

Yes, this is the chain rule for function composition from calculus.

The Sigma ($\sum$) on the 2nd line is because the neuron is connected to each of the neurons in the next layer, which can have more than one neurons.

Let's vectorize them now.

$$\frac{\partial L}{\partial Z^{(l+1)}} = \frac{\partial L}{\partial \begin{bmatrix} z_1^{(l+1)} \\ z_2^{(l+1)} \\ \vdots \\ z_{n^{(l+1)}}^{(l+1)} \end{bmatrix}_{n^{(l+1)} \times 1}}$$

$$= \begin{bmatrix} \frac{\partial L}{\partial z_1} \\ \frac{\partial L}{\partial z_2} \\ \vdots \\ \frac{\partial L}{\partial z_{n^{(l+1)}}} \end{bmatrix}_{n^{(l+1)} \times 1}$$

$$\frac{\partial Z^{(l+1)}}{\partial A^{(l)}} = \frac{\partial \begin{bmatrix} z_1^{(l+1)} \\ z_2^{(l+1)} \\ \vdots \\ z_{n^{(l+1)}}^{(l+1)} \end{bmatrix}_{n^{(l+1)} \times 1}}{\partial \begin{bmatrix} a_1^{(l)} \\ a_2^{(l)} \\ \vdots \\ a_{n^{(l)}}^{(l)} \end{bmatrix}_{n^{(l)} \times 1}}$$

$$= \begin{bmatrix} \frac{\partial z_1^{(l+1)}}{\partial a_1^{(l)}} & \frac{\partial z_1^{(l+1)}}{\partial a_2^{(l)}} & \cdots & \frac{\partial z_1^{(l+1)}}{\partial a_{n^{(l)}}^{(l)}} \\ \frac{\partial z_2^{(l+1)}}{\partial a_1^{(l)}} & \frac{\partial z_2^{(l+1)}}{\partial a_2^{(l)}} & \cdots & \frac{\partial z_2^{(l+1)}}{\partial a_{n^{(l)}}^{(l)}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial z_{n^{(l+1)}}^{(l+1)}}{\partial a_1^{(l)}} & \frac{\partial z_{n^{(l+1)}}^{(l+1)}}{\partial a_2^{(l)}} & \cdots & \frac{\partial z_{n^{(l+1)}}^{(l+1)}}{\partial a_{n^{(l)}}^{(l)}} \end{bmatrix}_{n^{(l+1)} \times n^{(l)}}$$

$$= \begin{bmatrix} w_{11}^{(l+1)} & w_{12}^{(l+1)} & \cdots & w_{1,n^{(l)}}^{(l+1)} \\ w_{21}^{(l+1)} & w_{22}^{(l+1)} & \cdots & w_{2,n^{(l)}}^{(l+1)} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n^{(l+1)},1}^{(l+1)} & w_{n^{(l+1)},2}^{(l+1)} & \cdots & w_{n^{(l+1)},n^{(l)}}^{(l+1)} \end{bmatrix}_{n^{(l+1)} \times n^{(l)}}$$

$$= W_{n^{(l+1)} \times n^{(l)}}^{(l+1)}$$

How do we combine $\frac{\partial L}{\partial Z^{(l+1)}}$ and $\frac{\partial Z^{(l+1)}}{\partial A^{(l)}}$? **We have to combine them with matrix operations in a way that is consistent with the underlying scaler operations**. Such a way is:

$$\left(\frac{\partial Z^{(l+1)}}{\partial A^{(l)}}\right)^T \cdot \frac{\partial L}{\partial Z^{(l+1)}}$$

, which gives a matrix of order $n^{(l)} \times 1$. To understand this, keep the orders of the factor matrices in mind, and check if what happens when you multiply is consistent with underlying scaler operations.

$$\frac{\partial A^{(l)}}{\partial Z^{(l)}} = \frac{\partial \begin{bmatrix} a_1^{(l)} \\ a_2^{(l)} \\ \vdots \\ a_{n^{(l)}}^{(l)} \end{bmatrix}_{n^{(l)}\times 1}}{\partial \begin{bmatrix} z_1^{(l)} \\ z_2^{(l)} \\ \vdots \\ z_{n^{(l)}}^{(l)} \end{bmatrix}_{n^{(l)}\times 1}}$$

$$= \begin{bmatrix} \frac{\partial a_1^{(l)}}{\partial z_1^{(l)}} & \frac{\partial a_1^{(l)}}{\partial z_2^{(l)}} & \cdots & \frac{\partial a_1^{(l)}}{\partial z_{n^{(l)}}^{(l)}} \\ \frac{\partial a_2^{(l)}}{\partial z_1^{(l)}} & \frac{\partial a_2^{(l)}}{\partial z_2^{(l)}} & \cdots & \frac{\partial a_2^{(l)}}{\partial z_{n^{(l)}}^{(l)}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial a_{n^{(l)}}^{(l)}}{\partial z_1^{(l)}} & \frac{\partial a_{n^{(l)}}^{(l)}}{\partial z_2^{(l)}} & \cdots & \frac{\partial a_{n^{(l)}}^{(l)}}{\partial z_{n^{(l)}}^{(l)}} \end{bmatrix}_{n^{(l)}\times n^{(l)}}$$

$$= \begin{bmatrix} \frac{\partial a_1^{(l)}}{\partial z_1^{(l)}} & 0 & \cdots & 0 \\ 0 & \frac{\partial a_2^{(l)}}{\partial z_2^{(l)}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\partial a_{n^{(l)}}^{(l)}}{\partial z_{n^{(l)}}^{(l)}} \end{bmatrix}_{n^{(l)}\times n^{(l)}}$$

, which is a diagonal matrix. We can combine it with incoming gradient this way:

$$\left(\frac{\partial Z^{(l+1)}}{\partial A^{(l)}}\right)^T \cdot \frac{\partial L}{\partial Z^{(l+1)}} \odot \operatorname{diag}\left(\frac{\partial A^{(l)}}{\partial Z^{(l)}}\right)$$

Yes, $\odot$ denotes point-wise multiplication of matrices, that is, **Hadamard Multiplication**. And, **diag(A)** is the diagonal vector of a diagonal matrix **A**. This combination give a matrix of order $n^{(l)} \times 1$.

$$\frac{\partial Z^{(l)}}{\partial W^{(l)}} = \frac{\partial \begin{bmatrix} z_1^{(l)} \\ z_2^{(l)} \\ \vdots \\ z_{n^{(l)}}^{(l)} \end{bmatrix}_{n^{(l)} \times 1}}{\partial \begin{bmatrix} w_{11}^{(l)} & w_{12}^{(l)} & \cdots & w_{1,n^{(l-1)}}^{(l)} \\ w_{21}^{(l)} & w_{22}^{(l)} & \cdots & w_{2,n^{(l-1)}}^{(l)} \\ \cdots & \cdots & \ddots & \vdots \\ w_{n^{(l)},1}^{(l)} & w_{n^{(l)},2}^{(l)} & \cdots & w_{n^{(l)},n^{(l-1)}}^{(l)} \end{bmatrix}_{n^{(l)} \times n^{(l-1)}}}$$

$$= \begin{bmatrix} \frac{\partial z_1^{(l)}}{\partial w_{11}^{(l)}} & \frac{\partial z_1^{(l)}}{\partial w_{12}^{(l)}} & \cdots & \frac{\partial z_1^{(l)}}{\partial w_{1,n^{(l-1)}}^{(l)}} \\ \frac{\partial z_2^{(l)}}{\partial w_{11}^{(l)}} & \frac{\partial z_2^{(l)}}{\partial w_{12}^{(l)}} & \cdots & \frac{\partial z_2^{(l)}}{\partial w_{1,n^{(l-1)}}^{(l)}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial z_{n^{(l)}}^{(l)}}{\partial w_{11}^{(l)}} & \frac{\partial z_{n^{(l)}}^{(l)}}{\partial w_{12}^{(l)}} & \cdots & \frac{\partial z_{n^{(l)}}^{(l)}}{\partial w_{1,n^{(l-1)}}^{(l)}} \end{bmatrix}_{n^{(l)} \times n^{(l-1)}}$$

$$= \begin{bmatrix} a_1^{(l)} & a_2^{(l)} & \cdots & a_{n^{(l-1)}}^{(l)} \\ a_1^{(l)} & a_2^{(l)} & \cdots & a_{n^{(l-1)}}^{(l)} \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{(l)} & a_2^{(l)} & \cdots & a_{n^{(l-1)}}^{(l)} \end{bmatrix}_{n^{(l)} \times n^{(l-1)}}$$

$$= \begin{bmatrix} a_1^{(l)} & a_2^{(l)} & \cdots & a_{n^{(l-1)}}^{(l)} \end{bmatrix}_{1 \times n^{(l-1)}}$$
$$\text{(Downcasted with no loss of information)}$$

$$= \left(A^{(l-1)}\right)^T$$

Yes, we have not consider the full Jacobian Matrix here.

Now we have,

$$
\begin{aligned}
\frac{\partial L}{\partial W^{(l)}} &= \left(\frac{\partial Z^{(l+1)}}{\partial A^{(l)}}\right)^T \cdot \frac{\partial L}{\partial Z^{(l+1)}} \odot \operatorname{diag}\left(\frac{\partial A^{(l)}}{\partial Z^{(l)}}\right) \cdot \frac{\partial Z^{(l)}}{\partial W^{(l)}} \\
&= \left(W^{(l+1)}\right)^T \cdot \frac{\partial L}{\partial Z^{(l+1)}} \odot \operatorname{diag}\left(\frac{\partial A^{(l)}}{\partial Z^{(l)}}\right) \cdot \left(A^{(l-1)}\right)^T \\
&= \frac{\partial L}{\partial Z^{(l)}} \cdot \left(A^{(l-1)}\right)^T
\end{aligned}
$$

Similiary,

$$
\begin{aligned}
\frac{\partial L}{\partial B^{(l)}} &= \frac{\partial Z^{(l)}}{\partial B^{(l)}} \cdot \frac{\partial L}{\partial Z^{(l)}} \\
&= I \cdot \frac{\partial L}{\partial Z^{(l)}} \\
&= \frac{\partial L}{\partial Z^{(l)}}
\end{aligned}
$$

Output Layer

$$
\begin{aligned}
\frac{\partial L}{\partial Z^{(4)}} &= \hat{Y} - Y \\
&\quad \text{(A consequence of Sigmoid Output Activation followed by BCE Loss)} \\
&= [0.558] - [1] \\
&= [-0.442]
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial W^{(4)}} &= \frac{\partial L}{\partial Z^{(4)}} \cdot \left(A^{(3)}\right)^T \\
&= [-0.442] \begin{bmatrix} 0.378 & 0.478 \end{bmatrix} \\
&= \begin{bmatrix} -0.167076 & -0.211276 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial B^{(4)}} &= \frac{\partial L}{\partial Z^{(4)}} \\
&= [-0.442]
\end{aligned}
$$

Hidden Layer 3

$$
\begin{aligned}
\frac{\partial L}{\partial Z^{(3)}} &= \left(W^{(4)}\right)^T \cdot \frac{\partial L}{\partial Z^{(4)}} \odot \operatorname{diag}\left(\frac{\partial A^{(3)}}{\partial Z^{(3)}}\right) \\
&= \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} [-0.442] \odot \text{ReLU Derivative}\left(Z^{(3)}\right) \\
&= \begin{bmatrix} -0.0442 \\ -0.0884 \end{bmatrix} \odot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} -0.0442 \\ -0.0884 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial W^{(3)}} &= \frac{\partial L}{\partial Z^{(3)}} \cdot \left(A^{(2)}\right)^T \\
&= \begin{bmatrix} -0.0442 \\ -0.0884 \end{bmatrix} \begin{bmatrix} 0.33 & 0.43 & 0.53 \end{bmatrix} \\
&= \begin{bmatrix} -0.014586 & -0.019006 & -0.023426 \\ -0.029172 & -0.038012 & -0.046852 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial B^{(3)}} &= \frac{\partial L}{\partial Z^{(3)}} \\
&= \begin{bmatrix} -0.0442 \\ -0.0884 \end{bmatrix}
\end{aligned}
$$

Hidden Layer 2

$$\begin{aligned}
\frac{\partial L}{\partial Z^{(2)}} &= \left(W^{(3)}\right)^T \cdot \frac{\partial L}{\partial Z^{(3)}} \odot \operatorname{diag}\left(\frac{\partial A^{(2)}}{\partial Z^{(2)}}\right) \\
&= \begin{bmatrix} 0.1 & 0.1 \\ 0.2 & 0.2 \\ 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} -0.0442 \\ -0.0884 \end{bmatrix} \odot \text{ReLU Derivative}\left(Z^{(2)}\right) \\
&= \begin{bmatrix} -0.01326 \\ -0.02652 \\ -0.03978 \end{bmatrix} \odot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} -0.01326 \\ -0.02652 \\ -0.03978 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial L}{\partial W^{(2)}} &= \frac{\partial L}{\partial Z^{(2)}} \cdot \left(A^{(1)}\right)^T \\
&= \begin{bmatrix} -0.01326 \\ -0.02652 \\ -0.03978 \end{bmatrix} \begin{bmatrix} 0.7 & 0.8 \end{bmatrix} \\
&= \begin{bmatrix} -0.009282 & -0.010608 \\ -0.018564 & -0.021216 \\ -0.027846 & -0.031824 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial L}{\partial B^{(2)}} &= \frac{\partial L}{\partial Z^{(2)}} \\
&= \begin{bmatrix} -0.01326 \\ -0.02652 \\ -0.03978 \end{bmatrix}
\end{aligned}$$

Hidden Layer 1

$$\begin{aligned}
\frac{\partial L}{\partial Z^{(1)}} &= \left(W^{(2)}\right)^T \cdot \frac{\partial L}{\partial Z^{(2)}} \odot \mathrm{diag}\left(\frac{\partial A^{(1)}}{\partial Z^{(1)}}\right) \\
&= \begin{bmatrix} 0.1 & 0.1 & 0.1 \\ 0.2 & 0.2 & 0.2 \end{bmatrix} \begin{bmatrix} -0.01326 \\ -0.02652 \\ -0.03978 \end{bmatrix} \odot \text{ReLU Derivative}\left(Z^{(1)}\right) \\
&= \begin{bmatrix} -0.007956 \\ -0.015912 \end{bmatrix} \odot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} -0.007956 \\ -0.015912 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial L}{\partial W^{(1)}} &= \frac{\partial L}{\partial Z^{(1)}} \cdot \left(A^{(0)}\right)^T \\
&= \begin{bmatrix} -0.007956 \\ -0.015912 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} -0.007956 & -0.007956 & -0.007956 \\ -0.015912 & -0.015912 & -0.015912 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial L}{\partial B^{(1)}} &= \frac{\partial L}{\partial Z^{(1)}} \\
&= \begin{bmatrix} -0.007956 \\ -0.015912 \end{bmatrix}
\end{aligned}$$

TASK 06. Perform One Parameter Update

Absolute values of the calculated gradients are in tenth and hundreth order of current weights and biases. I choose $\eta = 1$.

$$W_{\text{new}}^{(l)} = W_{\text{old}}^{(l)} - \frac{\partial L}{\partial W^{(l)}}$$

$$B_{\text{new}}^{(l)} = B_{\text{old}}^{(l)} - \frac{\partial L}{\partial B^{(l)}}$$

Hidden Layer 1

$$\begin{aligned} W^{(1)}_{\text{new}} &= \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix} - \begin{bmatrix} -0.007956 & -0.007956 & -0.007956 \\ -0.015912 & -0.015912 & -0.015912 \end{bmatrix} \\ &= \begin{bmatrix} 0.107956 & 0.207956 & 0.307956 \\ 0.115912 & 0.215912 & 0.315912 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} B^{(1)}_{\text{new}} &= \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} - \begin{bmatrix} -0.007956 \\ -0.015912 \end{bmatrix} \\ &= \begin{bmatrix} 0.107956 \\ 0.215912 \end{bmatrix} \end{aligned}$$

Hidden Layer 2

$$\begin{aligned} W^{(2)}_{\text{new}} &= \begin{bmatrix} 0.1 & 0.2 \\ 0.1 & 0.2 \\ 0.1 & 0.2 \end{bmatrix} - \begin{bmatrix} -0.009282 & -0.010608 \\ -0.018564 & -0.021216 \\ -0.027846 & -0.031824 \end{bmatrix} \\ &= \begin{bmatrix} 0.109282 & 0.210608 \\ 0.118564 & 0.221216 \\ 0.127846 & 0.231824 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} B^{(2)}_{\text{new}} &= \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} - \begin{bmatrix} -0.01326 \\ -0.02652 \\ -0.03978 \end{bmatrix} \\ &= \begin{bmatrix} 0.11326 \\ 0.22652 \\ 0.33978 \end{bmatrix} \end{aligned}$$

Hidden Layer 3

$$\begin{aligned} W^{(3)}_{\text{new}} &= \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 \end{bmatrix} - \begin{bmatrix} -0.014586 & -0.019006 & -0.023426 \\ -0.029172 & -0.038012 & -0.046852 \end{bmatrix} \\ &= \begin{bmatrix} 0.114586 & 0.219006 & 0.323426 \\ 0.129172 & 0.238012 & 0.346852 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} B^{(3)}_{\text{new}} &= \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} - \begin{bmatrix} -0.0442 \\ -0.0884 \end{bmatrix} \\ &= \begin{bmatrix} 0.1442 \\ 0.2884 \end{bmatrix} \end{aligned}$$

Output Layer

$$\begin{aligned} W_{\text{new}}^{(3)} &= [0.1 \quad 0.2] - [-0.167076 \quad -0.211276] \\ &= [0.267076 \quad 0.411276] \end{aligned}$$

$$\begin{aligned} B_{\text{new}}^{(3)} &= [0.1] - [-0.442] \\ &= [0.542] \end{aligned}$$

Note

Another Forward Propagation with new Weights and Bias would be great to verify if our loss is decreased or not. But I am tired.